FINANCIAL MARKET INSTRUMENTS (FMI)

Financial Derivatives – Special Products

Session 16: Securitisation and MBS, credit default swaps, the 2008 crisis, and market-linked notes.

Session plan:

Time	Segment
0:00-0:15	Warm-up and finance vocabulary
0:15-0:45	Securitization and mortgage-backed securities
0:45-1:10	Credit spreads and credit default swaps
1:10-1:35	The 2008 financial crisis
1:35-1:50	Market linked notes
1:50-2:00	Quiz, group exercise and wrap-up

Theme: every idea maps to reliability engineering. Watch for independent versus common-mode failure.

Terminologies

Term	Meaning	Engineering analogy
Principal	The amount lent	The nominal load
Interest	The price of borrowing	Rent on capital
Default	Borrower fails to repay	A component failure
Yield	Return earned; riskier means higher	A risk premium
Basis point	1/100 of 1%, so 200 bp = 2%	A fine tolerance unit
Derivative	Contract whose value depends on something else	A sensor reading that triggers a payout

Securitization and MBS: how it works

Securitization turns many slow, illiquid loans into securities that investors can buy and sell.

1. Banks lend

Banks and housing finance companies lend to thousands of borrowers.

2. Loans are pooled

The loans are sold to a separate legal entity (an SPV), off the bank's books.

3. Tranches

Securities are cut into senior, mezzanine and equity layers by who absorbs losses first.

4. Investors get EMIs

Borrowers' monthly payments pass through to investors, and the bank recovers cash to lend again.

Engineering view: tranching works like crumple zones. The equity layer takes the damage first, the senior layer last.

Discuss: why would a lender who sells every loan onward care less about borrower quality?

The securitization flow at a glance

Borrowers
Take loans and pay EMIs every month.

→

Lender
Originates the loans and usually services them.

→

SPV
Buys the pool and issues tranches by seniority.

→

Investors
Buy the securities and receive the cash flows.

Loans and risk move right. The lender passes the loans and their default risk to the SPV, and the SPV passes slices of it to investors.
Cash moves left, then right again. Investors' money funds the SPV's purchase, the lender gets cash to lend again, and borrowers' EMIs flow back through the servicer to investors.

Why securitize, and what can go wrong

Benefits

Liquidity: a 20-year loan becomes a security that can be sold in a day.
Capital recycling: the lender gets cash back and lends again, so the same capital funds many more homes.
Diversification: one investor holds a slice of thousands of borrowers, not one.
Choice: tranches let cautious and risk-seeking investors each buy the risk level they want.

Risks

Moral hazard: if the seller keeps nothing, it has less reason to check borrower quality.
Opacity: buyers cannot inspect thousands of loans, so they rely on ratings and models.
Correlation: diversification fails when one cause hits every loan.
Complexity: each extra layer makes the real risk harder to see.

Securitize: Why risk & return differ by tranche

Illustrative pool: ₹500 crore of loans at 9% earns ₹45 crore of interest a year. Payments flow down in strict order.

Order	Tranche	Size	Promised return	Paid a year
1st	Senior	₹400 cr	7.5%	₹30 cr
2nd	Mezzanine	₹70 cr	11%	₹7.7 cr
3rd	Equity	₹30 cr	Whatever is left	₹7.3 cr, about 24%

Read it like a pipe network: the senior pipe is filled first, so it is safest and earns least. The equity pipe is filled last, so it earns the most when all goes well and is the first to run dry when borrowers stop paying.

If borrowers pay less: pay ₹40 crore and equity gets only ₹2.3 crore. Pay ₹32 crore and mezzanine is short by ₹5.7 crore, with equity getting nothing.

Losses run in reverse: the equity tranche absorbs the first losses, then mezzanine, then senior. This is called subordination, and it is the main credit enhancement in a securitization.

Why tranching depends on independence

Independent defaults

Each of 1,000 loans has a 3% default chance, unrelated to the others.

Losses cluster near 3% of the pool, about ₹15 crore of ₹500 crore. The equity tranche (₹30 crore) absorbs it, and the senior tranche is untouched.

Common-mode failure

A nationwide price fall raises every borrower's default chance at once.

Losses can reach 30% of the pool, ₹150 crore. Equity and mezzanine are wiped out, and the senior tranche loses ₹50 crore.

Rule of thumb: Diversification protects against independent failures, not against a shared cause.

MBS in numbers and in practice

Follow one loan: a Bengaluru buyer borrows ₹60 lakh for 20 years at 9%. The EMI is about ₹54,000, so the lender collects about ₹1.3 crore over two decades. Sold into a pool, it gets ₹60 lakh back now and lends it again.

Now pool 1,000 such loans (₹500 crore): senior ₹400 cr, mezzanine ₹70 cr, equity ₹30 cr. Who loses what?

Scenario	Pool loss	Equity	Mezzanine	Senior
Normal defaults	₹25 cr	Loses ₹25 cr	Safe	Safe
Bad year	₹60 cr	Wiped out	Loses ₹30 cr	Safe
Nationwide crash	₹150 cr	Wiped out	Wiped out	Loses ₹50 cr (12.5%)

Example 1: US agency MBS

Ginnie Mae's guaranteed pools (first issued 1970) made 30-year mortgages cheap and widely available.

Example 2: India's PTCs

NBFCs sell vehicle and home loan pools to banks and mutual funds, recover cash, and keep lending.

Example 1: US agency MBS

What happens

1. A bank makes government-insured home loans, such as FHA or VA loans.
2. It pools them and issues securities guaranteed by Ginnie Mae, a US government agency. The first such pass-through came in 1970.
3. A servicer collects each monthly payment, keeps a small fee, and passes the interest and principal to investors.
4. The bank gets cash and makes new loans.

What students should notice

Credit risk is removed: the guarantee means investors are paid even if borrowers default.

A new risk appears, prepayment: when interest rates fall, homeowners refinance and investors get principal back early, then must reinvest at lower rates.

Engineering parallel: the cash flow is a variable-rate flow, like a tank whose outlet opens wider when conditions change.

Example 2 : India's PTC market

An illustrative deal

A vehicle-loan NBFC has lent ₹500 crore to truck owners across many states. It needs cash to keep lending.

It sells the loans to a trust. The trust issues pass-through certificates (PTCs) to banks, mutual funds and insurers.

Truck owners keep paying EMIs to the NBFC, which acts as servicer and passes the collections to the trust, and on to PTC holders.

Rating agencies rate each tranche, and credit enhancement such as a cash reserve or a junior tranche protects the senior investors.

Why it matters

For the lender: cash comes back in months instead of years, so one pool of capital funds several rounds of lending.
For investors: they get exposure to granular retail loans they could not originate themselves.
Post-2008 safeguard: RBI rules require originators to retain a share of the risk, so the seller still has skin in the game.

CHECKPOINT

Checkpoint: securitization

1. Why does the equity tranche earn the most?
2. Which risk replaces credit risk in a guaranteed agency MBS?
3. In the ₹500 crore pool, how much does the senior tranche lose if the pool loses ₹150 crore?

Credit Spreads and Credit Default Swaps

The mechanics

Credit spread = company yield minus government yield. 9% minus 7% = 200 basis points. A wider spread means the market fears default more.
CDS = a warranty on a loan. On ₹100 crore at 200 bp, the buyer pays ₹2 crore a year. If the spread jumps to 600 bp, the same cover costs ₹6 crore.
Why it matters: the spread is a live market sensor, and the CDS moves risk from one party to another.
Risks: anyone can buy one, the seller may not be able to pay, and pre-2008 trading was opaque.

Example 1: AIG

Sold protection on the top layers of mortgage-linked CDOs, collected premiums, and held almost no reserves. When prices fell it had to post cash it did not have. The US lent $85 billion on 16 September 2008, and total support exceeded $180 billion.

Example 2: Credit Suisse, 2023

Its CDS spread surged as confidence fell, before UBS took it over on 19 March 2023. The spread worked as an early-warning sensor, and the fear fed on itself.

Credit Spread

Rule of thumb: spread = chance of default × loss if it defaults. This is failure rate times consequence, the same logic as risk in safety engineering.

Observed spread	Loss if default	Implied yearly default chance
2% (200 bp)	60% (40% recovered)	2 ÷ 0.6, about 3.3%
6% (600 bp)	60%	6 ÷ 0.6 = 10%
0.5% (50 bp)	60%	0.5 ÷ 0.6, about 0.8%

What moves a spread: the borrower's earnings and debt, the economy, and investor fear. In a panic spreads widen even for sound borrowers because investors demand extra compensation for uncertainty.

Why it matters: a spread is a market-implied failure rate that updates daily. It is a sensor reading, not a forecast, and it includes a fear premium.

Credit Default Swap (CDS)

Protection buyer

Pays a fixed premium each quarter. Often a bank or fund worried about a borrower, or a trader betting on its decline.

Protection seller

Collects the premium and pays out if a credit event occurs. Often a large insurer or bank.

Reference entity

The company, country or security being insured. It is not a party to the contract and may not know it exists.

Credit events that trigger a payout include failure to pay, bankruptcy, and in many contracts a debt restructuring.

Settlement: the seller either pays face value minus the recovered amount in cash, or takes the defaulted bond and pays face value.

Two features that surprised people: a buyer needs no loss to collect (a "naked" CDS), and the seller's ability to pay is not guaranteed, so the insurance chain can break.

Example: CDS

Step	What happens	Numbers
1. Contract	A bank buys protection on a company's bonds	₹100 crore notional, 200 bp
2. Premium	The bank pays the seller every year	₹2 crore a year
3. Stress	The company's health worsens, so the spread widens	200 bp to 600 bp
4. Default	Bondholders recover only 40%	Loss of ₹60 crore
5. Payout	The seller pays the bank the loss	₹60 crore

Counterparty risk: if the seller cannot pay, the protection is worthless. That is what threatened AIG's buyers in 2008.

Carry the US loan forward: a bank holds a $100 million AAA layer of a CDO built from loans like the 2/28 loan and buys CDS at 0.5%, or $500,000 a year. If the layer loses 30% of its value, the seller owes $30 million. AIG sold this kind of cover with almost no reserves. (Figures are illustrative.)

Example 1: AIG's insurance machine

How it unfolded

1. AIG's financial products unit sold CDS protection on the top, supposedly safest, layers of mortgage-linked CDOs, collecting steady premiums.
2. It assumed these layers would almost never lose money and set aside very little to pay claims.
3. As house prices fell, the securities lost value and AIG was downgraded. The contracts then required it to post cash collateral to its buyers.
4. It could not find the cash. On 16 September 2008 the US Federal Reserve lent it $85 billion in return for about an 80% stake.

The lesson

Premiums looked like free money until the correlated event arrived. Selling insurance against a shared cause without reserves is a design with no safety factor.

Counterparty risk is systemic: banks that had bought protection from AIG were exposed to its failure.

Scale: total US rescue commitments to AIG exceeded $180 billion.

Engineering parallel: a warranty company insuring one product line against the same flaw, with no reserve fund. Collateral calls turned a paper loss into a cash crisis, so liquidity risk matters as much as credit risk.

Example 2: Credit Suisse, March 2023

How the signal moved

1. Confidence in the bank had been weak for years after repeated losses and scandals, so its CDS spread already sat well above its peers'.

2. In March 2023, the failure of US regional banks made investors nervous about weaker banks everywhere.

3. Credit Suisse's CDS spread jumped as its shares fell and depositors withdrew funds. The price of insuring its debt rose within days.

4. Swiss authorities arranged a takeover by UBS, announced on 19 March 2023.

What students should take away

The spread is a live sensor. It moved daily with news, while credit ratings changed far more slowly.
Fear can be self-reinforcing: a wider spread raises funding costs and spreads more worry, which widens the spread again. This is a positive feedback loop.
Not every loss was equal: some bondholders were wiped out in the rescue, which shows that even senior-sounding debt is only as safe as the structure around it.
Engineering parallel: a control loop with positive feedback runs away from its set point. A circuit breaker here is central bank liquidity support.

CHECKPOINT

Checkpoint: spreads and CDS

1. What does 600 bp cost a year to insure ₹100 crore?
2. With a 6% spread and 60% loss if default, what yearly default chance does the market imply?
3. Why can CDS protection fail even when you hold the contract?

CDOs: Securities made from securities

How it was built

1. Thousands of mortgages are pooled into an MBS and cut into tranches.
2. The senior layers sell easily. The middle (BBB-rated) layers are harder to sell.
3. Many of those middle layers from different MBS are pooled into a new pool, a CDO, and cut into tranches again.
4. Because the CDO's senior layer is protected by the layers below it, a large share of it was rated AAA, even though every piece inside was BBB.

Why it was fragile

Hidden assumption: the BBB pieces would fail independently. If each has a 5% chance of failing and failures are unrelated, losing more than a fifth of them at once is very unlikely.
Reality: all the pieces held loans exposed to the same national housing market, so they failed together.
Result: the "safe" senior CDO layers lost large amounts. A second layer of tranching multiplied the model's error.

The 2008 Financial Crisis: how it chained together

1. Cheap money

Low interest rates raise housing demand, so prices climb and everyone expects them to keep climbing.

2. Subprime loans

Weak borrowers get low "teaser" rates that reset higher after two years, assuming they can refinance.

3. Sell it onward

Lenders are paid per loan sold, so quality does not cost them. This is broken feedback.

4. MBS and CDOs

Loans are repackaged twice, and ratings assume independent failures, so many tranches get AAA.

5. Leverage and CDS

Banks borrow heavily to hold them, and insurers sell protection with thin reserves.

6. Prices fall

Resets hit, defaults rise together, and losses cannot be traced. Lehman fails on 15 September 2008.

The feedback loop: falling prices cause defaults, defaults force sales, forced sales push prices down further. Positive feedback drives the system away from stability.

2008 as a timeline

When	Event	Why it mattered
2006	US house prices peak and start to fall	The shared cause begins
Aug 2007	BNP Paribas freezes three funds exposed to US mortgages	First sign that nobody can value the securities
Sep 2007	Run on Northern Rock in the UK	Funding fear reaches depositors
Mar 2008	Bear Stearns rescued with official support	A major bank nearly fails
7 Sep 2008	Fannie Mae and Freddie Mac placed under US government control	The mortgage backbone is rescued
15-16 Sep 2008	Lehman files for bankruptcy; AIG rescued	Panic goes global
3 Oct 2008	US passes a $700 billion bank rescue program	Governments take the losses

2008 as a failure analysis

Failure type	Engineering parallel	2008 case
Model risk	Assuming independent failures	AAA ratings on tranches built on correlated loans
Hidden coupling	Unmapped dependencies	CDOs and CDS tied firms together invisibly
No fault isolation	A cascade with no circuit breaker	US subprime losses spread worldwide
Broken feedback	A supplier with no warranty cost	Originate-to-distribute lending
Thin safety factor	Designing with almost no margin	Banks with high leverage and little equity

2008 in two real examples

Example 1: One subprime loan, traced

A US family borrows $300,000 on a 2/28 adjustable loan: 6% for two years, then a reset. The payment is about $1,800 a month.

At the reset, an illustrative 9.5% rate lifts it to about $2,490, up 38%. With prices falling they cannot refinance, so they default.

Their loan sat in a pool backing an MBS, whose BBB layer went into a CDO, whose AAA layer AIG insured. One default is tiny. Millions in the same pools broke the chain.

Example 2: Lehman and the freeze

Lehman held mortgage assets funded by short-term borrowing. Lenders stopped rolling over its funding, and it filed for bankruptcy on 15 September 2008.

The next day a money market fund holding Lehman paper fell below $1 per unit. Short-term funding froze worldwide, and the Sensex fell from about 21,000 in January 2008 to about 8,000 by October.

What changed: Basel III capital rules, Dodd-Frank, risk retention for securitizers, and bank stress tests.

Lehman Brothers, in detail: the 72-hour cascade

Before the failure

Lehman held large positions in mortgage-related assets, financed mostly with borrowed money, at very high leverage.

Much of that borrowing was short-term, rolled over every few days. If lenders refused to roll it over, Lehman had to sell assets at falling prices.

As losses mounted, lenders did exactly that. A rescue deal collapsed over the weekend of 13-14 September 2008.

After the failure

15 September: Lehman files for bankruptcy, the largest in US history, with over $600 billion in assets.
16 September: a money market fund holding Lehman debt "breaks the buck", falling below $1 per unit. Investors rush to withdraw from similar funds.
Next days: short-term funding freezes, and companies struggle to pay routine bills. The shock reaches India as foreign investors sell.
Engineering parallel: a structure that stays up only while a support is continuously renewed. A run is a structural failure mode, and guarantees days later acted as isolation.

CHECKPOINT

Checkpoint: the 2008 crisis

1. Why did originate-to-distribute weaken lending standards?
2. Which assumption let CDO layers earn AAA ratings?
3. Why did Lehman's failure freeze short-term funding markets?

Market linked notes

A debt instrument whose return depends on a market variable such as the Nifty 50. In India they are market linked debentures. The issuer splits your ₹10 lakh for 5 years into two parts:

About ₹7.13 lakh: bond

A zero-coupon bond at 7% grows to ₹10 lakh ($10 \div 1.07^5$). This is the principal protection.

About ₹2.87 lakh: options

The remainder buys options that pay extra if the Nifty rises, and expire worthless if it falls.

The trade-off: you cannot have full upside and full protection, because the protection uses up most of the money. Protection also depends on the issuer paying.

Example 1: Nifty-linked debenture

At 70% participation, a 50% Nifty rise pays 35% over five years, about 6% a year simple. A fall returns principal, if the issuer pays.

Example 2: Lehman minibonds

Sold to retail investors in Hong Kong and Singapore as safe income. When Lehman failed, many were worth a fraction of face value.

How a market linked note is built and paid

Payoff formula: maturity value = principal × (1 + participation × max(0, index return))

Term	Meaning	Effect on you
Underlying	The index or basket the return tracks, such as the Nifty 50	Sets what you are betting on
Tenor	Length of the note, such as 5 years	Your money is locked in
Participation rate	Share of the gain you receive, such as 70%	Lower rate means less upside
Cap	Maximum return, if any	Limits your best case
Protection level	Share of principal returned if the market falls	Sets your worst case, if the issuer pays

Why participation is below 100%: the options that give you upside cost money. The more protection you buy, the less money is left for options, so the less upside you receive.

Market linked note payoff: ₹10 lakh, 5 years

Nifty after 5 years	Note return (70% participation)	Value at maturity
Up 50%	35%	₹13.5 lakh
Flat	0%	₹10 lakh
Down 20%	0%, principal protected	₹10 lakh
Issuer defaults	Protection fails	Part or all of the principal at risk

Compare: a 7% fixed return for five years grows ₹10 lakh to about ₹14.03 lakh, which beats the note even when the Nifty rises 50%.

Market linked notes: risks and a buyer's checklist

Risks

Issuer credit risk: protection fails if the issuer fails, as with Lehman.
Opportunity cost: capped or partial upside can trail a plain fixed deposit.
Illiquidity: selling before maturity may be hard or at a poor price.
Hidden fees: costs sit inside the structure rather than on a bill.
Complexity: payoff rules can hide how often you earn nothing.

Five questions before buying

1. Who is the issuer, and how strong is its credit?
2. Can I draw the payoff for a rise, a flat market and a fall?
3. What are the cap, participation rate and total fees?
4. Can I exit early, and at what price?
5. Would a simple alternative give a similar result?

Check your understanding

1. Why can a senior tranche lose money?
2. What does a rising CDS spread signal?
3. Name two reasons the 2008 losses spread worldwide.
4. What two parts make up a principal-protected note?
5. When is a note's "protection" not safe?

Five takeaways

1. Securitization slices risk, but it only works if loans fail independently.
2. A credit spread prices default risk, and a CDS lets you insure or trade it.
3. 2008 was a systems failure: correlated defaults, bad incentives, leverage, opacity, and flawed models.
4. A market linked note is a safe bond plus a derivative, so check the issuer.
5. Complexity hides risk. If you cannot draw the payoff and the failure modes, do not buy it.